STUN RUNNER (1989)

Complete Restoration Guide for Public Placements

Cabinet • Controls • Power • Monitor • Reliability

Prepared for **Coin Op 4 Charity** / Arcade Angels

Making Atari's high-speed tunnel racer reliable and inviting for youth and community spaces

1. Overview & Goals

S.T.U.N. Runner (Atari Games, 1989) is a fast, first-person tunnel racer known for its intense speed and distinctive sit-down or upright cabinets. For public and youth-center placements the priorities are reliability, safe and responsive controls, clean presentation, and a smooth player experience. This guide covers the whole machine so a restored unit can be placed with confidence.

2. Safety

- Unplug the cabinet before any internal work.
- CRT monitors store high voltage — discharge the anode before working on the chassis or neckboard.
- Steering and throttle mechanisms involve moving parts; keep hands clear when testing under power.
- If CRT work is outside your comfort zone, an LCD conversion is a practical option for high-use public sites.

3. Cabinet Exterior & Cosmetics

- Clean the entire cabinet thoroughly. Remove sticky residue, dust, and years of hand prints.
- Inspect side art, front decals, bezel, and any control-panel graphics. Replace or carefully restore heavily damaged art so the machine looks cared for.
- Marquee: clean the plastic and replace the fluorescent tube/starter if dim. A bright marquee makes the game far more inviting.
- Coin door: clean, lubricate, confirm locks or convert to free-play for youth placements and document the setting.
- Level the cabinet so it sits solidly — important for a high-speed driving / tunnel racer.

4. Steering, Throttle & Controls

These are the heart of the player experience on S.T.U.N. Runner.

- Steering: clean the wheel / yoke and column. Inspect the optical or potentiometer sensor. Clean contacts, check for worn bushings, and confirm smooth, accurate tracking with no dead zones or excessive play.
- Throttle / accelerator: clean and free any sticky action. Confirm the sensor or pot registers smoothly across its range and returns properly.
- Fire / weapon buttons: clean or replace sticky buttons and verify reliable switch action.

Sticky or inaccurate controls ruin the high-speed feel of this game — spend focused time here.

5. Power Supply & Edge Connector

Clean the power supply area thoroughly. Check and replace oxidized or burned fuse holders. Verify stable voltages at the board under load. The edge connector is a frequent source of intermittent boots or resets — clean the PCB fingers and the cabinet-side connector carefully. A modern switching supply conversion is a common reliability upgrade if the original supply is marginal.

6. Game Boards

Reseat boards firmly. Clean edge connectors. Look for obvious corrosion, broken traces, or heat-damaged components. If the machine has a history of random resets with power present, the edge connector and power path are the first places to check. Document any non-original boards or ROM sets for future volunteers.

7. Monitor

CRT path

Common issues include weak colors, focus problems, brightness drift, and chassis failures. Start with cleaning, checking monitor fuses, confirming isolation power, and reseating the neckboard. Only attempt deeper CRT work if experienced and following proper discharge procedures.

LCD conversion

An LCD conversion reduces heat, eliminates high-voltage hazards, and provides consistent picture quality with lower ongoing maintenance. Some cabinet modification or mounting work is usually required. Preserve the original CRT and chassis when possible.

8. Sound, Coin Door & Interior

- Confirm the speaker(s) produce clean engine, weapon, and effect sounds at reasonable volume. Replace if distorted.
- Clear dust from the interior, especially around the power supply and boards. Heat shortens component life.
- Confirm the rear interlock switch functions for safety.
- For free-play youth placements, set and document free-play or appropriate credit settings.

9. Final Testing Checklist Before Placement

- Full cold boot to attract mode with correct picture and sound
- Steering tracks smoothly and accurately with no dead zones
- Throttle / accelerator operates smoothly and returns properly
- Fire / weapon buttons register cleanly
- Complete several full runs with no freezes or resets
- Monitor picture is focused and bright enough from normal player distance
- Cabinet exterior is clean, level, and free of sharp edges or loose parts
- Marquee is lit and readable
- Free-play or pricing configured correctly for the host site
- Keys, spare fuses, and basic notes bagged and labeled for the host organization

10. Notes for Host Sites

Provide a short one-page sheet: power on/off, free-play status, and who to contact. Recommend a weekly wipe-down of the steering controls and glass. Ask the host to report any new stickiness, steering inaccuracy, or picture changes early. Document any non-original parts (LCD, switching supply, rebuilt controls, etc.) for future volunteers.

This guide is intended for educational and nonprofit restoration use by Coin Op 4 Charity and partner organizations. The goal is a S.T.U.N. Runner that stays reliable, safe, and fun for the people who play it.

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